Stock Price Forecasting and Risk Estimation

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Abstract – This study compares the accuracy of traditional statistical models like ARIMA with machine learning algorithms for predicting stock prices and evaluate their performance based on RMSE. Furthermore, we develop and evaluate advanced risk measures like VaR for AR/GARCH models using monthly log returns to predict one-step-ahead log returns as well as to implement portfolio optimization.
Index Terms – Stock, Time-series, Forecasting, Value-at-Risk

I. Introduction

Investors are always looking for ways to maximize their returns while minimizing their risks. Accurate prediction of future stock prices and estimation of risk measures are crucial for making informed investment decisions. This research aims to develop effective and robust models and techniques for stock price forecasting and risk measures estimation that can provide accurate and reliable predictions and help investors make informed decisions.

A. Problem Statement

Forecasting stock prices accurately is a challenging task due to the volatile and unpredictable nature of financial markets. While traditional statistical models like ARIMA have been used in the past, their accuracy is limited. Machine learning models have shown promise in recent years, but their performance varies depending on the data and methodology used. This research paper aims to develop and compare accurate and reliable models for stock price forecasting using both ARIMA and machine learning techniques. We will evaluate their performance based on metrics like MSE, MAE, and RMSE and identify their strengths and limitations in terms of forecasting accuracy, robustness, and computational efficiency.

Risk measurement is crucial for stock price forecasting in investment management. Various approaches like VaR and ES have been proposed to improve accuracy and robustness, but their application to AR/GARCH models is still an active area of research. In this study, we aim to develop and evaluate advanced risk measures for AR/GARCH models using monthly log returns to predict one-step-ahead log returns. We will estimate risk measures like VaR based on the return distribution and use them to adjust portfolio risk by increasing or decreasing exposure to specific assets. This research has significant implications for investors and financial analysts who rely on accurate and robust risk measures to make informed investment decisions and manage their portfolios effectively.

B. Contribution

The project on stock price forecasting and risk estimation has made significant contributions to the field of finance and investment. By utilizing advanced statistical models and machine learning algorithms, this project aims to predict the future performance of stocks and estimate the associated risks. The accurate forecasting of stock prices is crucial for investors to make informed decisions and maximize their returns.

Additionally, the risk estimation aspect of this project helps investors to assess the potential downside of their investments and make risk-adjusted decisions. Overall, the project on Stock Price Forecasting and Risk Estimation has the potential to revolutionize the investment landscape by providing valuable insights to investors and reducing the uncertainty and volatility associated with the stock market.

II. Methods

A. Dataset

In this research, we analyze three daily price Yahoo finance datasets Google, JPMorgan, and Tesla using time series and machine learning models. The datasets cover a decade, from 2012-01-01 to 2022-12-31, and use adjust price to analyzed using the R language. Below is the original time series of the three datasets.

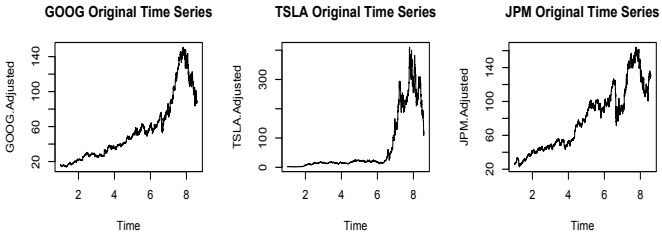


Fig 1. Different Original Time Series

B. Pre-processing

Detrending in time series analysis involves removing the trend component to isolate the residual component. Differencing or regression can be used for detrending. For forecasting with time series data, the ACF plot can help select optimal parameters for an ARIMA model. ARIMA models capture data dynamics through lagged observations,

differencing, and moving average terms. Optimal parameters are selected based on the ACF plot's characteristics, such as the lag where autocorrelation drops off or significant partial autocorrelation at certain lags.

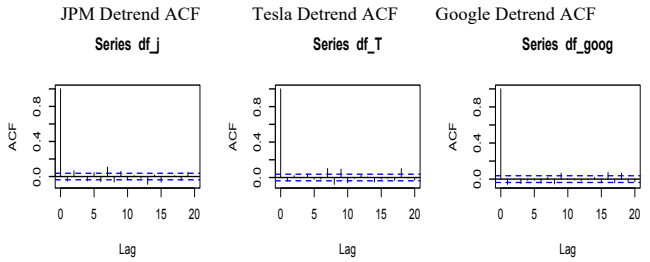


Fig 2. Different Detrend ACFs

For risk estimation, the monthly log return pertains to the logarithmic return on the percentage of income. That can potentially be gained or lost over a given month. Upon visual inspection of the data, a noticeable dip in returns is observed at the onset of the COVID pandemic. Furthermore, the variance in returns is distributed more widely, indicating greater variability in returns during this period. Notably, the fluctuations in returns for Tesla exhibit a significant divergence from the observed trend in other assets.

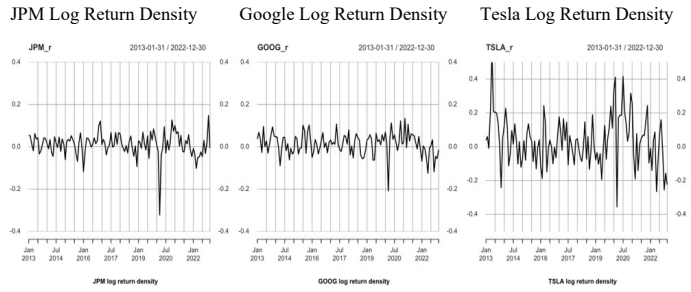


Fig 3: Log Return for JPM, GOOG and Tesla

The following graphs depict the log return density, showcasing a skewed normal distribution. Specifically, JPMorgan and Google exhibit a right-skewed distribution, while Tesla displays a left-skewed distribution.

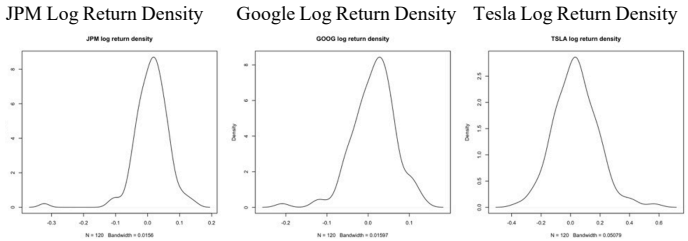


Fig 4: Different Distribution Curve as Log Return Density

The ACF of log returns can provide valuable insights for developing a GARCH model. Specifically, by examining the ACF plot, we can identify the presence of any significant autocorrelation in the data. If we observe a significant level of autocorrelation at certain lags, this indicates that the variance of the data is also likely to be correlated across those lags.

Based on these observations, we can construct a GARCH model that captures the time-varying volatility of the log returns. By incorporating the identified lags of autocorrelation into the GARCH model, we can generate a more accurate representation of the conditional variance of the log returns. This, in turn, can improve the reliability and effectiveness of any subsequent analyses or predictions based on the model.

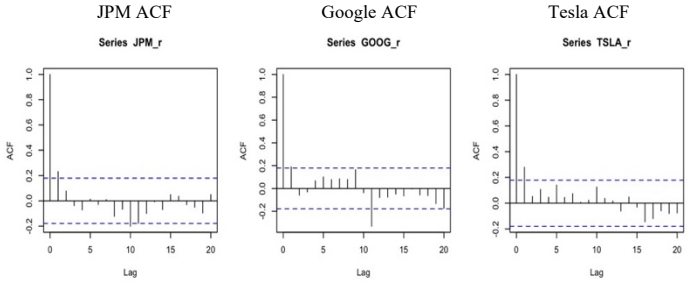


Fig 5. ACFs for stock log returns

C. *Algorithms*

The algorithm of the project consists of two parts: stock price forecasting and risk estimation. And the detailed algorithms and necessary explanation and formula will be explained below.

- Stock price forecasting

For stock price forecasting, an algorithm was set up before fitting the models. For ARIMA models, those parameters are estimated by using auto.arima() function in R. The optimal parameters for each stock are used for further analysis and the results are shown below.

```
Series: train_data
ARIMA(0,1,2)

Coefficients:
          ma1     ma2
      -0.0470  0.0700
s.e.   0.0189  0.0197

sigma^2 = 2.201:  log likelihood = -5016.43
AIC=10038.86   AICc=10038.87   BIC=10056.64
Series: train_data
ARIMA(1,1,1)

Coefficients:
         ar1      ma1
      0.7132  -0.7633
s.e.  0.1114   0.1028

sigma^2 = 1.483:  log likelihood = -4470.36
AIC=8946.72   AICc=8946.73   BIC=8964.5
Series: train_data
ARIMA(0,1,1)

Coefficients:
          ma1
      -0.0361
s.e.   0.0186

sigma^2 = 22.1:  log likelihood = -8208.66
AIC=16421.33   AICc=16421.33   BIC=16433.18
```

Fig 6. Estimated parameters of ARIMA models

As you can see from the tables above, ARIMA (0,1,2), ARIMA (1,1,1), and ARIMA (0,1,1) is fitted for the JMP(Chase), Google, and Tesla stock, respectively.

For Random Forest model, a few independent variables need to be created before fitting the random forest because time series data does not have any variables except the target variable, but random forest requires the variables. To do so, 'lag 1' and 'lag 2' of the time series data are created added as a column to the raw data as independent variables taking first and second lags to the dataset. There is a case for JPM(Chase) shown below.

```
             adj_JPM     lag1     lag2
2012-01-05 26.04306 25.51023 25.34965
2012-01-06 25.80949 26.04306 25.51023
2012-01-09 25.76570 25.80949 26.04306
2012-01-10 26.31312 25.76570 25.80949
2012-01-11 26.75837 26.31312 25.76570
2012-01-12 26.89705 26.75837 26.31312
2012-01-13 26.21824 26.89705 26.75837
2012-01-17 25.48103 26.21824 26.89705
2012-01-18 26.67079 25.48103 26.21824
model <- randomForest(adj_JPM ~ lag1 + lag2, data = train, ntree = 1000)
```

Fig 7. An example of creating two lags as independent variables

The lags are passed to the randomForest() function for each company as the JPM(Chase) shown above. Then, the new variables ('lag 1' and 'lag 2') are used to fit the random forest model, with the parameter 'ntree' of 1,000. Those fitted models are also evaluated and compared in further analysis.

- Risk estimation

For risk estimation, an AR-GARCH model was fitted to monthly log return data. The aim was to predict the distribution of one-step-ahead log returns distribution, which would enable the calculation of risk, with a focus on the Value-at-Risk.

The reason behind incorporating a GARCH component in the model instead of using a pure AR model is because stock returns exhibit time-varying volatility in the market. This implies that the magnitude of returns can differ significantly over time. An AR model, however, does not consider this feature and assumes a constant variance of the errors over time.

In contrast, an AR-GARCH model captures both the autocorrelation in the returns and the volatility clustering that is commonly observed in financial time series data, resulting in more reliable risk estimates. The formula of the AR-GARCH model is in:

$$X_{t+1} = \mu + \phi X_t + e_{t+1}$$
$$e_{t+1} = \sigma_{t+1} + \varepsilon_{t+1}$$
$$\sigma_{t+1}^2 = \omega + \alpha e_t^2 + \beta \sigma_t^2$$

It can be observed that X_{t+1} closely resembles an AR model, with the exception that the noise term, e, assumes not constant and consists of two components. The symbol σ indicates the time-varying variability of uncertainty or volatility, and ε refers to the standardized error term that we can simulate through copula modeling subsequently.

In the study, an AR (1)-GARCH (1,1) model was initially applied to estimate the residuals from the fitted model. A residual analysis was subsequently conducted to assess the model's capability to capture serial correlation and heteroscedasticity. The diagnostic tests employed included the ACF plot, Normal Q-Q plot, and Box-Ljung test, which were utilized to evaluate the normality of the residuals and detect the presence of serial correlation. Additionally, the Breusch-Pagan test was employed to confirm the presence of heteroscedasticity in the residuals. Utilizing these approaches can help determine the optimal order for the AR component of the model.

Following the identification of the optimal model, copula models were employed to model the dependence between residuals, specifically the estimated ε. Copulas are statistical techniques that enable the joint distribution of two or more random variables to be modeled, while also modeling their dependence structure independently of their marginal distributions. By fitting a copula model to the residuals obtained from the AR-GARCH model, the dependence structure between residuals can be better captured, resulting in enhanced forecasting and risk management strategies. In addition, various copula algorithms were tested, and the algorithm with the best performance was ultimately selected.

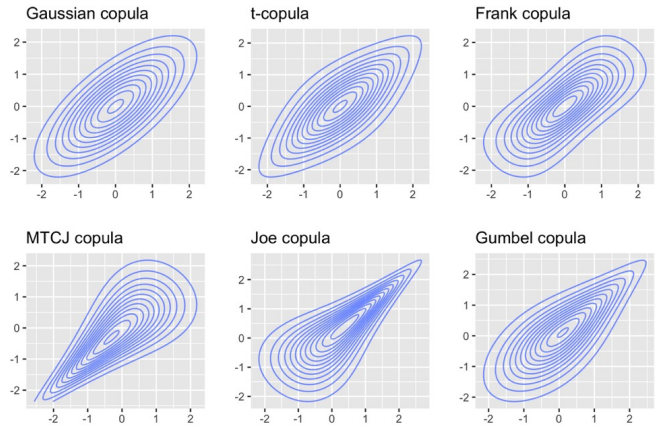


Fig. 8 Different copulas algorithms

Therefore, considering the parameters provided in (X) and the error term obtained from the copula simulation, we can numerically calculate the one-step-ahead return of X_{n+1} using the formula (X). Then, after obtaining the distribution of X_{n+1} returns, which includes all potential return scenarios of the stocks, we can estimate the probability as well as the magnitude of the future losses using the Value-at-Risk measure.

Value-at-Risk (VaR) is a financial risk management tool that provides an estimation of potential losses in the value of an investment or portfolio within a specified time horizon and level of confidence. It represents the maximum monetary amount that an investor is willing to lose under normal market conditions. In the context of this project, VaR will be expressed as a percentage value indicating the portion of initial investment that is likely to be lost over the next month.

Finally, having obtained the computed one-step-ahead return distribution and Value at Risk (VaR) for the targeted stocks, we want to delve into the realm of portfolio optimization. This

entails maximizing the anticipated return while simultaneously minimizing the associated risk. Our study is delimited to the selection of two stocks within each portfolio, as we aim to ascertain the optimal allocation of weights for each stock.

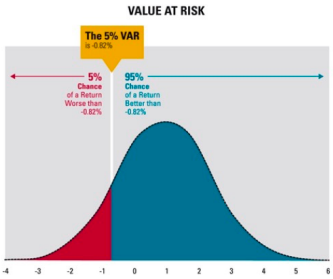


Fig. 9 Definition of Value-at-Risk

III. Results

A. Findings

- Stock price forecasting

For stock price forecasting, the performance of the models is measured based on the three companies, and then the two models, ARIMA and Random Forest, are compared as well.



Fig #. Comparison for actual and predicted values of ARIMA models

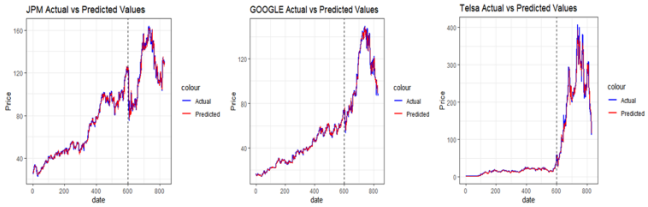


Fig. 10 Comparison for actual and predicted values of Random Forest models

Company	Measure	Value
JPM	MAE	0.9147
	MSE	2.1982
	RMSE	1.4826
Google	MAE	**0.6900**
	MSE	**1.4813**
	RMSE	**1.2171**
Tesla	MAE	1.8918
	MSE	22.0877
	RMSE	4.7000

Table 1. Result for three error measures of ARIMA models

Random Forest

Company	Measure	Value
JPM	MAE	1.0760
	MSE	2.7412
	RMSE	1.6557
Google	MAE	**0.8045**
	MSE	**1.8300**
	RMSE	**1.3528**
Tesla	MAE	2.2521
	MSE	31.3196
	RMSE	5.5964

Table 2. Result for three error measures of Random Forest models

Based on the results above, the fitted ARIMA model well predicts all three company's stocks. Especially, the model shows the best predictive performance with Google stock and the worst performance with Tesla. Similarly, the fitted Random Forest model also well predicts all three company's stocks, and the model shows the best predictive performance with Google stock and the worst performance with Tesla as the ARIMA model as well.

- Risk estimation

In the experiment, our study focused on analyzing JPM and GOOG stock performance to determine the best weights for each stock over the next month. We evaluated the models' performance and predicted the one-step-ahead log return distribution of both stocks. We also conducted portfolio optimization, specifically bootstrapping, to determine optimal weights based on factors such as expected return, volatility, and correlation between the stocks. The findings in risk estimation mainly include the models' performance report, prediction of the one-step-ahead log return distribution of the two stocks, portfolio optimization.

According to the results of the residual analysis, we have tested various orders for the autoregressive (AR) component and identified the optimal models for JPM and GOOG to be AR (3,0)-GARCH (1,1) and AR (2,0)-GARCH (1,1), respectively. The detailed coefficient report is presented.

```
Mean and Variance Equation:
 data ~ arma(3, 0) + garch(1, 1)
<environment: 0x12038f9c8>
 [data = JPM_r]

Conditional Distribution:
 std

Coefficient(s):
        mu         ar1         ar2         ar3       omega      alpha1       beta1       shape
 0.01145321  0.15375288  0.05091057 -0.04234073  0.00248659  0.00000001  0.00000001  4.26653155

Std. Errors:
 based on Hessian

Error Analysis:
        Estimate  Std. Error  t value Pr(>|t|)
mu     1.145e-02   4.236e-03    2.704  0.00686 **
ar1    1.538e-01   6.783e-02    2.267  0.02341 *
ar2    5.091e-02   7.166e-02    0.710  0.47742
ar3   -4.234e-02   8.913e-02   -0.475  0.63476
omega  2.487e-03   3.341e-03    0.744  0.45669
alpha1 1.000e-08         NaN      NaN      NaN
beta1  1.000e-08   1.397e+00    0.000  1.00000
shape  4.267e+00   1.519e+00    2.808  0.00498 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Fig. 11 Parameters from AR-GARCH model on JPM stock

```
Final Estimate of the Negative LLH:
 LLH:  -192.076    norm LLH:  -1.600633
          mu           ar1           ar2         omega        alpha1         beta1         shape
 1.333839e-02  2.026956e-01 -8.218028e-02  2.731479e-09  3.131750e-03  1.000000e+00  4.654795e+00

R-optimhess Difference Approximated Hessian Matrix:
                  mu           ar1           ar2         omega        alpha1         beta1         shape
mu      -54789.33813 -9.112877e+02 -8.291504e+02 -5.947322e+05  4.136317e+01 -1.324137e+03 -1.160148e+01
ar1       -911.28773 -1.654522e+02 -4.655514e+01 -4.646333e+04 -8.603506e+01 -1.383292e+02 -4.886529e-01
ar2       -829.15044 -4.655514e+01 -1.365662e+02 -2.020532e+05 -5.249660e+02 -5.975588e+02 -8.586917e-01
omega  -594732.17276 -4.646333e+04 -2.020532e+05 -1.918826e+10 -4.056307e+07 -5.168039e+07 -6.073595e+04
alpha1      41.36317 -8.603506e+01 -5.249660e+02 -4.056307e+07 -8.760549e+04 -1.101740e+05 -1.233301e+02
beta1    -1324.13742 -1.383292e+02 -5.975588e+02 -5.168039e+07 -1.101740e+05 -1.401770e+05 -1.682085e+02
shape      -11.60148 -4.886529e-01 -8.586917e-01 -6.073595e+04 -1.233301e+02 -1.682085e+02 -4.334246e-01
attr(,"time")
Time difference of 0.008944035 secs
```

Fig. 12 Parameters from AR-GARCH model on Google stock

Using the estimated parameters obtained from the time-series model and the standardized error terms from the copula model, we can calculate the next-month stock log return distribution, as illustrated in the figure. Our analysis reveals that the generated future log returns conform approximately to a normal distribution. This finding is highly significant as it supports the common assumption made in numerous financial models that the returns of financial assets, including stocks, bonds, and currencies, follow a normal distribution or at least closely approximate one. Such an assumption is predicated on the observation that financial asset returns are often found to exhibit a normal distribution or a distribution that is sufficiently close to normal.

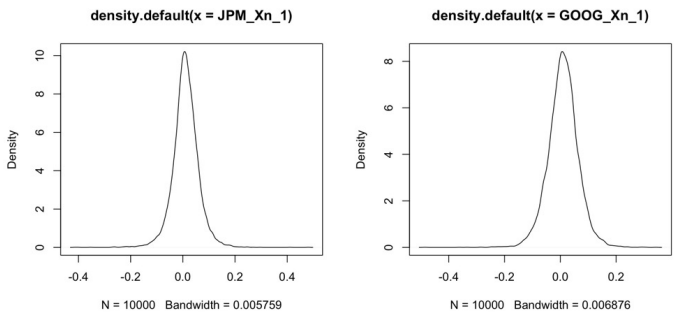


Fig. 13 Distribution of log return for JPM and GOOG

Then, given distributions, we can calculate the risk of the portfolio by changing the weights of each stock. The table and figure of bootstrapping will present this. From the result, we can see that the optimal portfolio with least risk includes 60% JPM stock and 40% GOOG stock. Thus, we can say there is 99% confidence that the portfolio will not lose more than 10.8% of the investment over next month.

	JPM	GOOG	VaR
1	0.1	0.9	0.1239
2	0.2	0.8	0.1192
3	0.3	0.7	0.1154
4	0.4	0.6	0.1123
5	0.5	0.5	0.1098
6	0.6	0.4	0.1077
7	0,7	0.3	0.1120
8	0.8	0.2	0.1140
9	0.9	0.1	0.1187

Table. 2 Portfolio composition and risk value VaR

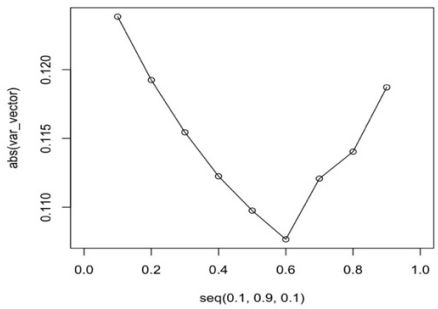


Fig. 14 Portfolio composition and VaR illustration

B. Evaluation

- Stock price forecasting

One of the goals that we wanted to see is comparison of predictive performance of ARIMA model Random Forest. In this sense, RMSE is selected as the measure for the performance since MSE basically shows the same conclusion with different scale from the original and RMSE is considered as more robust than MAE. The result table is shown below.

Company	ARIMA	Random Forest
	RMSE	
JPM	**1.4826**	1.6557
Google	**1.2171**	1.3528
Tesla	**4.7000**	5.5964

Fig. 15 Comparison for ARIMA and Random Forest models in terms of predictive performance

Throughout the three companies, the ARIMA model has a better *testRMSE*, therefore, we can conclude that ARIMA model is recommended to analyze stock price. The result can be easily acceptable because ARIMA models are designed specifically for time series data, on the other hand, Random Forest models are a more general-purpose machine learning algorithm.

- Risk estimation

In this study, we employed various statistical techniques to evaluate the performance of our model, specifically conducting residual analysis to assess normality, serial correlation, and heteroskedasticity. To accomplish this, we employed methods such as the ACF, Q-Q norm, Box-Ljung test, and Breusch-Pagan test.

Our results indicate that the residuals exhibit no significant autocorrelation and follow a normal distribution. Additionally, we found that the models constructed using data from JPM and GOOG both passed the Box-Ljung and Breusch-Pagan tests, indicating that they are well-equipped to capture serial correlation and heteroskedasticity. Based on these analyses, we conclude that the model is well-fitted to the data and provides valuable insights for further study.

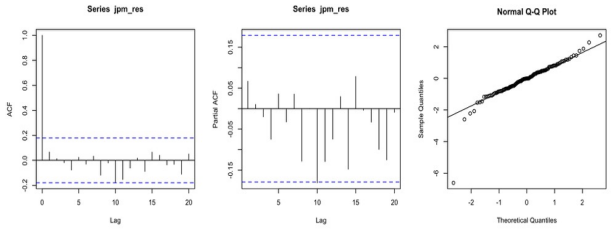


Fig. 16 Diagnostic plots for JPM

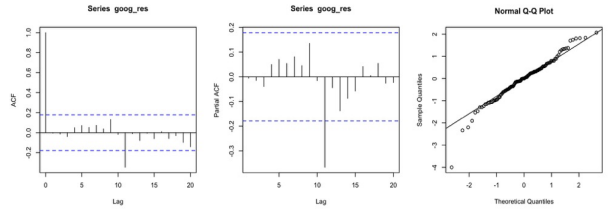


Fig. 17 Diagnostic plots for GOOG

```
        Box-Ljung test

data:  goog_res
X-squared = 4.7642, df = 8, p-value = 0.7825
```

Fig. 18 Box-Ljung test

```
        studentized Breusch-Pagan test

data:  goog_res ~ fitted(goog_fit)^2
BP = 1.1295, df = 1, p-value = 0.2879
```

Fig. 19 Breusch-Pagan tests

IV. Summary

A. Conclusion

The use of ARIMA and Random Forest models for stock price forecasting and VaR for risk estimation has made significant contributions to the finance and investment field. Our study shows that the ARIMA model outperformed the Random Forest model in terms of RMSE, making it a more reliable option for stock price forecasting. VaR proved to be an effective tool for estimating potential losses within a specific confidence interval, enabling investors to construct portfolios with a desired level of risk and return. However, it is important to note that risk measures are only estimates and may not accurately reflect future performance, and multiple measures should be used for thorough analysis before making investment decisions.

There are also limitations to the use of these models, such as the assumption of normality and stationarity in the ARIMA model and the potential overfitting of the Random Forest model. These limitations should be carefully considered when interpreting the results and making investment decisions. The findings of this research provide valuable insights to investors and contribute to the advancement of the finance and investment field. Accurate stock price forecasting and risk estimation are crucial for informed decision-making and maximizing returns.

The use of advanced statistical models and machine learning algorithms, such as the ARIMA and Random Forest models, and the VaR approach can revolutionize the investment landscape by providing valuable insights to investors and reducing the uncertainty and volatility associated with the stock market. Future research can focus on improving the accuracy and reliability of these models and exploring other methods for stock price forecasting and risk estimation.

B. Discussion

Future research can focus on expanding the scope of this project to include a broader range of stocks and other financial assets. Additionally, the incorporation of other factors such as news sentiment analysis and macroeconomic indicators can further improve the accuracy of the forecasting and risk estimation models. It is also essential to develop more sophisticated models to account for sudden changes in the market, such as unexpected events or shifts in investor sentiment. Furthermore, the project can be improved by providing investors with more comprehensive risk measures that consider other aspects of risk, such as liquidity and credit risk. Additionally, the project can benefit from the integration of ethical considerations in the investment decision-making process, such as environmental, social, and governance (ESG) factors.

In conclusion, the stock price forecasting and risk Estimation project has the potential to significantly contribute to the field of finance and investment. The continued development and improvement of these models can enhance the accuracy and reliability of stock price forecasting and risk estimation and provide valuable insights to investors.

V. Teamwork

In the investigation into stock price forecasting and risk estimation, our team has benefited significantly from the individual contributions of its members.

Ashley and Jongtaek have been pivotal in stock price forecasting, their expertise and acumen proving invaluable. Additionally, they have made valuable inputs during our team discussions, providing us with insightful guidance that has informed our decision-making processes.

On the other hand, Huaqian and Jiawei have spearheaded the development of our risk estimation model, undertaking extensive research into financial risk metrics and methodologies. Moreover, they have been instrumental in conducting comprehensive back testing and validation of our models, ensuring their accuracy and reliability.

It is worth noting that each member of our team has played a crucial role in the success of our research, and we take immense pride in the collaborative effort we have put forth.